Swati Aggrawal

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PROFESSIONAL SUMMARY

M.Sc. Computer Science graduate from Sapienza University of Rome with a strong foundation in software engineering, cloud-based deployment, and applied AI/ML. Experienced in developing AI and computer vision solutions using Python, PyTorch, TensorFlow, FastAPI, Docker, and AWS. Conducted research on Vision Transformers and continual learning, with an IEEE publication and hands-on experience building scalable machine learning applications. Passionate about applying AI to solve real-world industrial challenges.

SKILLS

- **Programming Languages:** Python, C++, Java, SQL, Bash
- **AI & Machine Learning:** Machine Learning, Deep Learning, Computer Vision, Vision Transformers, Continual Learning, Transfer Learning, PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face Transformers, Generative AI, LLMs, Prompt Engineering, RAG, AI Agents
- **Software Engineering:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), FastAPI, REST APIs, Git
- **Cloud & DevOps:** Docker, Cloud Deployment (AWS ECS, Fargate, CloudWatch), Azure, Linux
- **AI-Assisted Development:** GitHub Copilot, Claude —leveraged AI-assisted development tools for code generation, debugging, documentation, and rapid prototyping.
- **Data Science & Visualization:** NumPy, Pandas, Matplotlib

EDUCATION

Sapienza Università di Roma ***May 2026***
Master of Science in Computer Science

- Relevant Coursework: Natural Language Processing, Deep Learning and Applied AI, Machine Learning, Computer Vision, Distributed Systems
- Thesis: Evaluating Pretrained Vision Transformers Under Continual Learning.
- Developed and evaluated adapter-based continual learning methods for Vision Transformers using PyTorch and Elastic Weight Consolidation (EWC) to mitigate catastrophic forgetting.

Sharda University ***July 2023***
Bachelor of Technology – CSE with AI and ML

- Relevant Coursework: Operating Systems, Data Structures, Machine Learning, Databases, Cloud Computing
- Thesis: Virtual Generative Model Inducing Human Identity to Computers in Metaverse
- Awarded merit-based scholarship (Top 5–10% of cohort).

EXPERIENCE

Researcher — Vision Lab, Sapienza Università di Roma (Master's Thesis) **Nov 2025 – May 2026**
github.com/swatiaggrawal/Thesis-Vision-Transformers

- Conducted research under Prof. Marco Raoul Marini on continual learning for computer vision in the Vision Lab.
- Developed and optimized PyTorch training and evaluation pipelines for continual learning experiments on Vision Transformers, benchmarking accuracy, forgetting, and computational efficiency across sequential classification tasks.
- Designed and implemented adapter-based learning and Elastic Weight Consolidation (EWC) techniques, achieving 90.59% average accuracy with only 2.01% average forgetting on Split CIFAR-100 (5 sequential tasks) — a ~97% relative reduction in catastrophic forgetting versus a CNN baseline.
- Benchmarked against ResNet-18 and naive ViT fine-tuning baselines (24.22% and 41.91% average accuracy respectively), validating results via cross-seed evaluation.
- Collaborated with researchers to document findings and compare state-of-the-art continual learning approaches.
- Technologies: Python, PyTorch, Vision Transformers, Computer Vision

PUBLICATION

Machine Learning for Air Pollution — IEEE Conference Publication (2023)
doi.org/10.1109/ViTECoN58111.2023.10157028

- Evaluated Random Forest, SVM, Decision Tree, and Neural Network models on environmental datasets.
- Performed comparative analysis using accuracy, robustness, and generalization metrics.
- Authored and presented findings through an IEEE peer-reviewed publication.

PROJECTS

DocDelta — AI-Powered Document Intelligence & Comparison Platform
github.com/swatiaggrawal/DocDelta

- Developed an end-to-end semantic document comparison API using FastAPI and Sentence Transformers (all-MiniLM-L6-v2), detecting and classifying sentence-level changes (added/removed/modified, numerical vs. wording) beyond traditional character-level diffs.
- Designed scalable embedding and cosine-similarity retrieval pipelines, achieving ~840ms warm-request latency on CPU-only inference.
- Containerized the application with Docker and deployed on AWS ECS/Fargate with CloudWatch monitoring for latency and error tracking.
- Built REST APIs with async request handling and validation, optimizing container size (CPU-only PyTorch builds) and eliminating cold-start overhead via model preloading.
- Technologies: Python, FastAPI, Sentence Transformers, PyTorch, Docker, AWS ECS Fargate, CloudWatch, REST APIs

Metal Defect Detection using Computer Vision
github.com/swatiaggrawal/Metal-Defect-Detection

- Built a custom CNN (TensorFlow/Keras) plus SVM and Random Forest classifiers (HOG features) for automated industrial metal surface defect detection across 6 defect classes on the NEU dataset.
- Built image preprocessing, augmentation, training, and evaluation pipelines, tuning classical models via GridSearchCV.
- Achieved 97.22% test accuracy with the CNN versus 86.11% for both SVM and Random Forest — an ~11-point improvement demonstrating the value of learned spatial features over hand-crafted descriptors.
- Evaluated models using accuracy, confusion matrices, precision, recall, and F1-score.
- Technologies: Python, TensorFlow, Keras, OpenCV, Scikit-Learn

Lossy Text Compressor [Semantic NLP Pipeline]
github.com/swatiaggrawal/LossyTextCompressor

- Developed a semantic text compression pipeline using GTR-T5 embeddings, N-bit vector quantization, and Vec2Text beam-search reconstruction.
- Evaluated reconstruction quality across 8 quantization levels (2-16 bit) using cosine similarity, BLEU, and ROUGE-L metrics.
- Identified an optimal 6-bit quantization sweet spot (0.96 cosine similarity, 0.78 ROUGE-L) and a sharp quality collapse beyond 8-bit, revealing a phase transition in embedding-space distortion.
- Technologies: Python, PyTorch, Transformer Models, NLP, Sentence Embeddings